

The Mathematics of the Claim Graph: Formal Structure and Computational Capacity

Executive Summary

The physics-kernel epistemic state presents itself as a finite directed acyclic multigraph of typed claims, but structurally, it operates as a mathematical hybrid. No single established formalism maps perfectly to its complete architecture without generating structural friction. Rigorous analysis of the mathematical literature identifies the object as a discrete, time-indexed family of AND-OR directed multigraphs, fibered over an independent provenance graph, and evaluated via a monotone fixpoint over a bounded, partially incomparable lattice.

Three formalisms provide the strongest exact tools for this specific architectural hybrid. First, the framework of Provenance Semirings—specifically Absorptive Polynomials (Sorp)—perfectly mirrors the AND-OR grounding and the lattice-based fixpoint evaluation, providing an exact, theorem-backed algebra for evidence composition. Second, Monotone Boolean Circuit theory maps grounding directly to circuit evaluation, where circuit complexity theorems yield exact algorithms for identifying minimal sufficient evidence via prime implicants. Third, Discrete Hodge Theory (HodgeRank) and Cellular Sheaves offer an algebraic-topological approach, allowing the contradiction subgraph to be analyzed as an edge flow that cleanly separates localized logical friction from global epistemic frustration.

The sharpest precondition failure discovered in this analysis relates to the continuous metrics of gradual argumentation and opinion dynamics. The kernel utilizes a finite ordinal lattice with two partially incomparable modes and per-kind ceilings. However, the majority of the literature on weighted bipolar argumentation and continuous opinion dynamics on sheaves assumes totally ordered cardinal values (e.g., real numbers in $[0, 1]$ or real vector spaces). Consequently, axiomatic guarantees in gradual argumentation—particularly those involving compensation or continuous limits—are fundamentally broken when applied to the kernel. To maintain mathematical rigor, aggregation operators must be strictly constrained to the domain of product lattices or coalesced sums, utilizing ordinal measurement theory rather than cardinal arithmetic.

The Object: Formalisms and Structural Isomorphisms

The mathematical literature offers several formalisms that approximate the kernel's epistemic state. Because the kernel relies on exact computation at a small scale (currently order two hundred nodes, scaling to the low thousands), asymptotic bounds and probabilistic approximations are less relevant than exact structural homomorphisms. The following table identifies the formal match, the structural mismatch, and the single most useful theorem each framework contributes.

Formalism	Exact Match	Precondition Mismatch	Most Useful Theorem / Contribution
Monotone Boolean Circuits	AND-OR grounding groups map directly to	The grade lattice is non-Boolean; circuit	Theorem: The prime implicants of a

Formalism	Exact Match	Precondition Mismatch	Most Useful Theorem / Contribution
	conjunction/disjunction gates; acyclic dependency maps to feedforward circuit topology.	formalism natively supports $\{0, 1\}$ or continuous reals, lacking multi-modal ordinality.	monotone Boolean function uniquely define the minimal sufficient evidence sets (the exact minimal support footprint).
Provenance Semirings	Tracks exact evidence propagation. The absorptive semiring $\text{Sorp}(X)$ supports exact tracking of recursive derivations through fixpoints.	The kernel's grade lattice is an ordered monoid, not necessarily a full semiring, complicating distributive law guarantees.	Theorem: Any query evaluation over an absorptive, fully continuous semiring commutes with the least fixpoint operator, guaranteeing finite convergence.
Bipolar Abstract Argumentation	The directed multigraph of support and contradiction maps natively to bipolar abstract argumentation frameworks.	Gradual semantics literature relies heavily on continuous weight intervals $[0, 1]$, violating the discrete, incomparable dual-mode lattice.	Theorem: Representation theorems explicitly relate the equivalence principle to specific impact and aggregation operator structural conditions.
Assumption-Based TMS (ATMS)	Natively tracks minimal environments (support sets) under which nodes hold; explicitly accommodates and isolates contradictions.	ATMS exact computation scales at worst-case exponential; it tracks total propositional logic rather than just monotone paths.	Theorem: A one-to-one correspondence exists between admissible extensions of a truth maintenance system and the minimal environments of the ATMS.
Formal Concept Analysis (FCA)	The claim-evidence incidence defines a formal context; the Galois connection derives concept hierarchies automatically.	FCA assumes a flat bipartite object-attribute matrix, whereas the kernel features deep, transitive DAG dependencies.	Theorem: The fundamental theorem of concept lattices proves that Galois connection fixed points form a complete lattice representing the implicit taxonomy.
Curry-Howard / Proof Nets	Claims equate to types; support edges equate to proof terms; acyclic verification equates to type checking.	Pure Curry-Howard does not support graded truth, contradictory edges, or probabilistic degradation natively.	Theorem: Cut elimination on proof hypergraphs provides an exact structural procedure for simplifying transitive

Formalism	Exact Match	Precondition Mismatch	Most Useful Theorem / Contribution
			dependencies without information loss.

Monotone Boolean Circuits and AND-OR Graphs

Viewing the grounding structure as a monotone Boolean circuit provides profound computational insights. In this isomorphism, claims act as gates, where conjunctive support groups correspond to AND gates, and alternative support groups correspond to OR gates. Theoretical computer science demonstrates that the exact minimal support sets for a claim correspond precisely to the prime implicants of the monotone function computed at that claim's node. A claim's prime implicants represent the minimal sufficient evidence required to maintain its standing. While standard circuit complexity typically operates over the Boolean domain $\{0, 1\}$, the structural derivation of prime implicants remains valid for ordinal structures provided the operations remain strictly monotone. The primary precondition failure arises from the kernel's incomparable modes; a pure Boolean circuit evaluates a single truth dimension, whereas the kernel evaluates a composite lattice.

Provenance Semirings and Absorptive Polynomials

The database theory community formalized the tracking of derivation structures using provenance semirings, an approach spearheaded by Green, Karvounarakis, and Tannen. By assigning variables (provenance tokens) to base facts and tracking them through algebraic operations—where multiplication represents joint use (AND) and addition represents alternative use (OR)—the system builds provenance polynomials. For the physics-kernel, the specific instantiation of the *absorptive polynomial semiring* (Sorp) is the exact mathematical match. An absorptive semiring enforces the axiom $a + ab = a$, which elegantly captures the logical redundancy of evidence: if claim A is sufficient, the combination of claim A and claim B provides no strictly greater logical footprint. Literature confirms that computations in $\text{Sorp}(X)$ or its fully continuous completion $S^\infty[X]$ permit the exact calculation of both least and greatest fixpoints without infinite expansion, perfectly mirroring the kernel's monotone fixpoint over evidence.

Bipolar Argumentation and Gradual Semantics

Abstract argumentation, originating with Dung, models epistemic states as graphs of attacks and supports. Weighted bipolar argumentation extends this by assigning initial weights to arguments and propagating them through gradual semantics. The kernel's structure is closely related to a bipolar weighted argumentation framework, wherein the grade lattice represents the arguments' strength. However, the gradual semantics literature is dominated by continuous mathematical formulations, mapping strengths to the real interval $[0, 1]$. Axiomatic properties designed for continuous spaces, such as compensation (where multiple weak attacks algebraically counterbalance one strong support), mathematically fail on the kernel's discrete ordinal lattice. The absence of continuous limits renders the standard iterative convergence proofs of bilateral gradual semantics invalid for the kernel, requiring instead lattice-theoretic discrete fixpoint guarantees.

Justification Logic and the ATMS

The Assumption-Based Truth Maintenance System (ATMS), developed by de Kleer, explicitly maintains the minimal environments (footprints) under which a node holds true. The ATMS operates by tracking labels—sets of minimal assumptions—and propagating them through justification networks. This is mathematically identical to calculating the prime implicants of a monotone circuit, but optimized for artificial intelligence reasoning engines. The ATMS natively handles contradictions by declaring specific assumption sets as "nogoods," which aligns perfectly with the kernel's preservation of contradictory edges as held states. While the ATMS suffers from exponential worst-case scaling on large knowledge bases, at the kernel's scale ($N \approx 200$), the computation is exact, instantaneous, and provides the definitive footprint-closure algorithm.

Formal Concept Analysis (FCA)

Formal Concept Analysis uses a Galois connection to map a bipartite graph of objects and attributes into a hierarchical concept lattice. If claims are viewed as objects and their supporting evidence footprint as attributes, FCA automatically generates a rigid, complete taxonomy of the epistemic state. However, FCA assumes a flat incidence matrix. The kernel's dependency structure is a deep DAG, meaning the attribute mapping must be transitively closed before FCA can be applied. While mathematically valid, mapping a deep DAG to a flat incidence matrix often results in a combinatorial explosion of the concept lattice, obscuring the human-readable hierarchy.

Curry-Howard Isomorphism and Proof Nets

Through the Curry-Howard correspondence, claims serve as types, and the support edges serve as proof terms. The gate evaluation acts as a type-checking algorithm. Linear logic and proof nets provide graphical representations of these proofs, utilizing cut elimination to simplify transitive dependencies. The primary contribution of this perspective is structural: proof relevance dictates that the exact topology of the support matters, not just the boolean reachability. However, the classical Curry-Howard formulation assumes absolute truth and binary type inhabitation, failing to accommodate the kernel's probabilistic degradation, finite lattice grades, and maintained contradiction edges without highly speculative extensions into graded dependent type theory.

Verification Analysis, Theorem-Backed

The kernel requires deterministic diagnostic tools to evaluate the fragility, dependency, and consistency of the epistemic state. At a scale of two hundred claims, graph algorithms traditionally dismissed as computationally intractable become viable, allowing exact reads of the network's structural integrity.

Single Points of Failure (AND-OR Cuts)

In standard directed graphs, single points of failure are identified via articulation vertices or minimal cut sets. However, applying standard graph connectivity theorems to an AND-OR

structure yields mathematically invalid results. A claim's failure only disconnects downstream nodes if it is a critical input to an AND group, or the terminal remaining input to an OR group. This maps exactly to identifying minimal vertex cuts in a directed hypergraph, where hyperedges represent the conjunctive AND-groups. Recent mathematical literature on hypergraph splitting-off operations provides local connectivity-preserving reductions. By converting the AND-OR graph into a monotone Boolean circuit, a single point of failure is defined as any node whose zero-evaluation collapses the root node to zero. *Verdict:* CONFIRMED (Verification Analysis). *Cost:* Polynomial time via circuit evaluation. *Maturity:* THEOREM.

Independence as Disjointness (Menger's Theorem)

To verify that two audits or evidence chains are independent, their provenance footprints must be entirely disjoint. Menger's Theorem states that the maximum number of vertex-disjoint paths equals the minimum vertex cut. However, Menger's theorem on standard graphs cannot process the required simultaneous satisfaction of AND-groups. The correct generalization lies in directed hypergraphs, where finding maximum disjoint hyperpaths is generally NP-hard. Fortunately, at the kernel's scale, approximation is unnecessary. Independence is exactly computed by evaluating the intersection of the ATMS minimal environments (prime implicants) of the target nodes. If the intersection of their minimal support sets is empty, the claims are mathematically independent. *Verdict:* CONFIRMED (Verification Analysis). *Cost:* Exact and cheap via prime implicant intersection. *Maturity:* THEOREM.

Contamination Dynamics and Percolation

When a claim is withdrawn, the contamination cascades through the dependency structure. Network science models this via k-core decomposition, robustness analysis, and percolation on directed graphs. However, these mathematical tools rely upon thermodynamic limits and asymptotic approximations designed for graphs with millions of nodes. Applying statistical percolation theory to a two-hundred-node deterministic logic graph is mathematically vacuous and epistemically misleading. At this scale, the withdrawal cascade is strictly governed by the exact recalculation of the monotone fixpoint. *Verdict:* DEAD (Precondition Failure). *Cost:* N/A. *Maturity:* SPECULATIVE (for this scale).

Sensitivity Ranking (Shapley Values)

To determine which single grade change exerts the highest influence on the downstream epistemic state, one must calculate exact influence on a monotone circuit. This aligns precisely with the computation of Banzhaf and Shapley indices. In the database provenance literature, Livshits, Bertossi, Kimelfeld, and Senellart have extensively analyzed the Shapley value of database facts. While computing Shapley values over arbitrary Boolean expressions is FP-#P-complete, their research demonstrates that when the provenance polynomial is compiled into a deterministic decomposable circuit (Knowledge Compilation), the Shapley value can be computed tractably. Because the kernel's state is small, the exact Shapley value of each support claim can be computed dynamically, yielding a mathematically rigorous, game-theoretic sensitivity ranking. *Verdict:* CONFIRMED (Verification Analysis). *Cost:* Tractable via Knowledge Compilation. *Maturity:* RESULT.

Minimal Inconsistent Sets (Conflict Cores)

Contradictions in the kernel are represented as held edges. Finding the core of a dispute maps directly to identifying Minimal Unsatisfiable Subsets (MUS) in SAT solving, or computing "conflict cores" in argumentation theory. Because the grade lattice evaluates these paths deterministically, the problem reduces to enumerating the prime implicants of the contradiction gate itself. Using the ATMS label propagation algorithm, the minimal inconsistent environments can be isolated exactly, highlighting the specific subsets of base claims causing the top-level contradiction. *Verdict*: CONFIRMED (Verification Analysis). *Cost*: Exact, fast at N=200. *Maturity*: THEOREM.

Audit Allocation

Allocating verification effort—such as determining which claims require manual review to secure a target theorem—is a facility location or set cover problem over the DAG. Given the small N, the exact optimal covering set can be calculated via integer linear programming (ILP) without needing to rely on the logarithmic approximation guarantees of greedy algorithms. *Verdict*: PLAUSIBLE (Verification Analysis). *Cost*: NP-hard in general, but trivial via standard ILP solvers at N=200. *Maturity*: APPLIED.

Synthesis Potential: Structure Pointing at Content

Synthesis operations ask where the mathematical structure of the graph implies missing knowledge, latent variables, or necessary architectural reorganization.

Link Prediction and Structural Holes

In sociological network theory, Ronald Burt's concept of structural holes identifies absent ties between dense clusters. Applied to the claim graph, two dense clusters of evidence with no bridging claims represent a candidate synthesis site (e.g., a missing theoretical unification or an undiscovered dependency). Measuring betweenness centrality or applying standard link-prediction algorithms over the support topology can automatically highlight these voids, providing a purely structural signal for literature-based discovery. *Verdict*: PLAUSIBLE (Synthesis Potential). *Maturity*: APPLIED.

Community Detection and Module Structure

Standard network analysis utilizes modularity maximization or flow-based clustering to detect latent communities. In the physics-kernel, applying flow-based clustering to the support DAG attempts to recover the latent semantic architecture. If the mathematically detected communities diverge significantly from the user-designed tier structure, the divergence itself is a synthesis finding. It indicates that the true epistemic dependencies cross the designed architectural boundaries, pointing toward necessary refactoring. *Verdict*: CONFIRMED (Synthesis Potential). *Maturity*: APPLIED.

Hodge Decomposition on Graphs (HodgeRank)

The kernel's contradiction edges, combined with the grades of the connected claims, form a structured pairwise-comparison dataset. Combinatorial Hodge theory (HodgeRank), developed by Jiang, Lim, Yao, and Ye, applies the Hodge Laplacian to discrete graphs. It decomposes edge flows into three orthogonal components:

1. **Gradient Flow:** The globally consistent ranking of claims (the resolving hierarchy).
2. **Harmonic Flow:** Global cyclic inconsistencies (e.g., a macro-level cycle where A contradicts B, B contradicts C, and C contradicts A).
3. **Curl Flow:** Localized triangular inconsistencies (noise). Applying HodgeRank to the contradiction subgraph enables the kernel to localize inconsistency algebraically. By separating the gradient from the harmonic and curl flows, the system can distinguish between localized, fixable disagreements (curl) and systemic, global frustration (harmonic). *Verdict:* CONFIRMED (Verification Analysis & Synthesis Potential). *Maturity:* APPLIED.

Sheaf-Theoretic Consistency

A more advanced topological tool is the cellular sheaf, which assigns vector spaces to vertices (stalks) and edges, with linear restriction maps dictating how data flows across the network. Hansen and Ghrist developed "discourse sheaves" to model opinion dynamics, allowing agents to hold private multi-dimensional opinions while projecting selectively into shared discourse spaces. In the kernel, a claim's grade and provenance could theoretically be modeled as a section of a cellular sheaf. The sheaf Laplacian then computes the obstruction to global consistency (sheaf cohomology). While theoretically elegant and capable of handling complex, heterogeneous data structures, this framework requires mapping the discrete grade lattice into inner product vector spaces. This violates the ordinal nature of the kernel's lattice, forcing artificial linearizations that distort the epistemic data. *Verdict:* NEEDS REFINEMENT (Synthesis Potential). *Maturity:* SPECULATIVE.

Spectral Methods

Analyzing the spectrum of the support DAG's Laplacian or non-backtracking operator can reveal structural bottlenecks and expansion properties. However, spectral methods rely heavily on the continuous relaxation of discrete graph properties, and their most powerful theorems are asymptotic. At $N=200$, the eigenvalues provide limited actionable synthesis data compared to exact combinatorial methods. *Verdict:* MARGINAL (Synthesis Potential). *Maturity:* THEOREM (but contextually weak).

Dependency-Structure from Real Corpora

Recent empirical analyses of the Lean Mathlib and Isabelle Archive of Formal Proofs (AFP) dependency graphs provide direct, mathematically rigorous baselines for typed claim graphs. Research on the TheoremGraph dataset demonstrates that formalized mathematical networks exhibit specific, measurable structural signatures: highly skewed (weakly scale-free) in-degree distributions (power-law exponent $\alpha \approx 1.81$), massive transitive redundancy (up to 17.5%), and deep critical paths (DAG depth of 153 for Mathlib's 308,129 declarations). These metrics are directly portable to the kernel. By comparing the physics-kernel's degree distribution and depth to the Mathlib baseline, one can predict refactor points and identify missing abstraction layers. Nodes exhibiting anomalously high in-degree centrality relative to the

baseline serve as implicit "bottleneck lemmas" requiring architectural elevation. *Verdict*: CONFIRMED (Synthesis Potential). *Maturity*: APPLIED.

The Grade Lattice as an Algebraic Object

The crux of the kernel's computation is the earned-grade evaluation, defined as a deterministic monotone fixpoint over a finite lattice with per-kind ceilings and two partially incomparable modes. The mathematical mechanics of this lattice dictate which operations are valid and which theorems transfer.

Domain Theory and Fixpoint Iteration

The Knaster-Tarski theorem guarantees that any monotone function on a complete lattice possesses a least and greatest fixpoint. Because the kernel's grade lattice is finite, Kleene iteration is guaranteed to reach this fixpoint in a finite, explicitly bounded number of steps. The abstract interpretation literature frames this mechanism precisely: the earned grade is an abstract interpretation of the underlying concrete evidence graph, formally connected by a Galois connection (where the lower adjoint abstracts the evidence into a grade, and the upper adjoint concretizes the grade into its minimal evidence requirement). The lack of infinite ascending chains in a small, finite lattice means that convergence accelerators like "widening" or "narrowing" operators—common in continuous domain theory—are mathematically irrelevant here.

The Challenge of Incomparable Modes

The prompt notes that the lattice contains two partially incomparable modes. In lattice theory, combining two distinct ordered sets can be done via a Cartesian product (maintaining independent dimensions) or a coalesced sum (gluing them at the bottom/top elements). If the incomparable modes represent fundamentally different types of epistemic status (e.g., "empirical verification" versus "theoretical consistency"), forcing them into a single total order destroys critical information. The aggregation functions over a product lattice must process the modes component-wise. If the system attempts to aggregate across the incomparable modes, the result will frequently collapse to the supremum ($\top$) or infimum ($\bot$), causing a catastrophic loss of precision. The lattice must be rigorously maintained as a product structure to prevent cross-modal contamination.

Ordinal vs. Cardinal Standing

The gradual argumentation literature frequently utilizes aggregation functions (such as division, exponentiation, or parameterized norms) that require cardinal numbers, treating weights as continuous variables in the real interval $[0,1]$. However, according to measurement theory, operations on an ordinal lattice are strictly limited to $\min$ (meet), $\max$ (join), and monotone step functions. Applying cardinal-based axiomatic principles—such as the Compensation principle in bilateral gradual semantics, which allows multiple weak attacks to arithmetically counterbalance one strong support—is mathematically invalid on an ordinal lattice. The kernel must rely strictly on lattice-theoretic polynomials (combinations of AND/min and OR/max).

Probabilistic Refinements

Frameworks such as Dempster-Shafer theory, imprecise probability, and Subjective Logic introduce explicit trust operators and interval bounds to handle uncertainty. While mature in sensor fusion and Bayesian network topologies, transitioning the kernel from an ordinal lattice to a probabilistic semiring requires shifting from $\text{Sorp}(X)$ to continuous tensor operations. This significantly increases computational complexity and violates the discrete, exact nature of the current architecture without necessarily yielding greater epistemic clarity for a human-in-the-loop verification system. *Verdict on Probabilistic Refinement*: SPECULATIVE (for this specific architecture).

The Free Experiments

The following four experiments are exact, computationally cheap at $N \approx 200$, implementable using standard structural libraries (e.g., NetworkX, SciPy), and falsifiable. They serve as immediate, theorem-backed affordance tests for the kernel's substrate.

Experiment 1: Exact Shapley Sensitivity Ranking

- **Computation:** Treat the graph's target claim as a Boolean function of its base evidence claims. Compute the exact Shapley value for every leaf node (source claim) using the Sorp provenance semiring equations via a decomposed circuit evaluation.
- **Literature Instantiation:** Database provenance Shapley values (Livshits, Bertossi, Kimelfeld, Senellart).
- **Expected Output:** A ranked list of leaf claims, weighted by their marginal contribution to the target claim's final grade.
- **Falsifiability:** The expected outcome is that highly cited base claims rank highest. The finding is *surprising* (and highly actionable) if an obscure, lowly cited claim holds a disproportionate Shapley mass due to its hidden position as a single point of failure in a critical AND-group.

Experiment 2: HodgeRank Decomposition of Contradictions

- **Computation:** Extract the contradiction subgraph. Assign edge flows based on the ordinal difference in the grades of the contradicting claims. Compute the discrete graph Laplacian and project the flow onto the exact gradient, harmonic, and curl subspaces.
- **Literature Instantiation:** Combinatorial Hodge Theory / HodgeRank (Jiang, Lim, Yao, Ye).
- **Expected Output:** A quantitative decomposition metric showing the percentage of conflict that is gradient (a resolvable hierarchy) versus harmonic (global cyclic frustration) versus curl (local triangular noise).
- **Falsifiability:** If the contradiction subgraph is purely gradient, the epistemic state is consistent and logically resolving. The finding is *surprising* if a significant harmonic flow exists, proving mathematically that the system harbors unresolvable, cyclic ideological deadlocks that require human architectural intervention.

Experiment 3: ATMS Prime Implicant (Minimal Environment)

Extraction

- **Computation:** Run a label-propagation algorithm (Boolean multiplication/addition) from the leaves to the root to calculate the exact sum-of-products (DNF) for a top-level claim. Minimize the polynomial using Boolean absorption ($a + ab = a$).
- **Literature Instantiation:** de Kleer's Assumption-Based Truth Maintenance System (ATMS); Provenance absorptive polynomials.
- **Expected Output:** A set of prime implicants (the minimal subsets of leaf claims necessary and sufficient to prove the root claim).
- **Falsifiability:** The expected outcome is that minimal environments respect designed modular boundaries. The finding is *surprising* if the prime implicants expose hidden, cross-disciplinary dependencies that bypass the intended semantic structure of the framework, revealing fragile theoretical bottlenecks.

Experiment 4: Mathlib Baseline Structural Comparison

- **Computation:** Calculate the in-degree distribution, modularity, and maximum DAG depth of the physics-kernel support graph. Compare these specific scalars directly to the extracted TheoremGraph dataset metrics for formal mathematics libraries.
- **Literature Instantiation:** The Network Structure of Mathlib; Archive of Formal Proofs complex network analysis.
- **Expected Output:** A tabular comparison of network topology metrics against the formal baseline.
- **Falsifiability:** The expected outcome is that the kernel graph exhibits shallow, broad topologies indicative of early-stage, informal modeling. The finding is *surprising* if the kernel already mimics the deep, scale-free, high-transitive-redundancy signatures of mature formalized mathematics, pointing directly to specific nodes that serve as implicit, highly loaded "bottleneck lemmas."

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